

Uday Sonawane

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<https://github.com/Udaydsmls> | <https://udaydsmls.github.io/>

EDUCATION

Northeastern University, Khoury College of Computer Sciences, Boston, MA

May 2026

Master of Science, Computer Science

- GPA 4.0/4.0
- Relevant Coursework: *Programming Design Paradigm, Cloud Computing, Foundations of AI, Algorithms, DBMS, ML*

Vellore Institute of Technology, Vellore, TN, India

April 2023

Bachelor of Technology, Computer Science

- GPA 8.55/10
- Relevant Coursework: *Software Engineering, Mathematical Model for Data Science, Operating Systems, Parallel Computing*

EXPERIENCE

Siemens Digital Industries Software, Pune, MH, India

June 2023 – Jan 2024

Software Engineer Intern

- **Enhanced Code Performance:** Improved system efficiency by fixing over 70% of errors through targeted enhancements in ReactJS and C++.
- **Boosted Product Stability:** Strengthened product reliability by engaging in Agile Release Train (ART) scrums and collaborating with cross-functional teams to identify and resolve bugs.
- **Streamlined Testing & Analysis:** Elevated code coverage and troubleshooting efficiency by developing automation tests and a Python regex tool for log error extraction, integrating with Polarion and CI/CD pipelines.

TATA AutoComp Systems Ltd, Pune, MH, India

May 2022 – June 2022

Data Science Intern

- **Improved battery-life predictions:** Developed neural networks and ensemble models to analyze driving-behavior telemetry, achieving up to 85% predictive accuracy and enabling more reliable battery life estimates.
- **Enhanced fault detection:** Implemented anomaly-detection pipelines using Isolation Forest and K-Means clustering to identify sensor faults and improve detection reliability across telemetry streams.
- **Delivered actionable insights:** Translated complex analyses and model outcomes into clear, prioritized recommendations and visualizations for user, enabling data-driven performance optimizations.

PROJECTS

[StockVision – AI Stock Prediction & Visualization](#) | Python, REST API, TensorFlow, C++, React.js, Full Stack | August 2025

- **Full-Stack ML Deployment:** Built an end-to-end stock-prediction platform: trained ML/DL models (LSTM, ARIMA, Gradient Boosting) in Python/TensorFlow, implemented high-performance inference in a C++ backend, and developed a React frontend for real-time visualization and interaction.
- **Predictive Modeling & Benchmarking:** Designed, trained, and benchmarked multiple forecasting approaches on Yahoo Finance datasets to evaluate accuracy and robustness across investment horizons; iterated on features and promoted top-performing models to production.

[VideoFrameRateEnhancer](#) | Python, TensorFlow, AI, Neural Networks, GPU Optimizations | February 2025

- **AI-Driven Frame Interpolation:** Built a CNN/CNN-LSTM model with Transposed CNN to generate intermediate frames, doubling frame rates (up to 4x improvement iteratively) and enhancing video smoothness.
- **Optimized Processing & Memory:** Streamlined GPU data flow and memory management through batch processing and dynamic RAM clearance, reducing processing bottlenecks by 40% for large-scale video data.

[SimpleRayTracer](#) | C++ | November 2024

- **Custom Ray Tracing Engine:** Built a ray tracing engine solely with the C++ Standard Library, applying linear algebra (vector operations, dot/cross products, and matrix transformations) to simulate accurate global illumination, shadows, reflections, and transparency.
- **Performance Optimizations & Dynamic Composition:** Developed efficient ray-object intersection algorithms with spatial partitioning and implemented a perspective camera featuring adjustable field of view, depth of field, and focus for dynamic scene composition.

SKILLS

Languages: C++, Python, Java, JavaScript, TypeScript, SQL, R, PHP, C

Frameworks & Libraries: AWS, React.js, Flask, Django, TensorFlow, PyTorch, OpenCV, NumPy, Pandas, scikit-learn

Technical Skills: AI, Machine Learning, Deep Learning, Data Science, Data Analytics, Software Development, GPU Optimizations, Computer Vision, Problem Solving, Github, Object-Oriented Programming, Cloud Technologies

Software Engineering: Software Design Patterns, Multithreading, RESTful APIs, Profiling & Debugging, Software Development Life Cycle

Certifications: Google Data Analytics Specialization, UC San Diego: Algorithmic Toolbox, Hackerrank: Problem Solving Intermediate

ACHIEVEMENTS

- Highest rating of 1439 at Codeforces, **ranked 502 out of 23k** in Contest (Round 944), which is the top site in terms of participation of programmers across globe (Problem Solving Skill).